

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS:

Total Marks: 10

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

- A. `<media src="video.mp4"></media>`
B. `<video src="video.mp4" controls></video>`

- C. <movie src="video.mp4"></movie>
- D. <vid src="video.mp4"></vid>

Snippet 2: Form Submission Methods

- A. GET
- B. POST
- C. PUT
- D. DELETE

Snippet 3: Navigation Elements

- A. <div>
- B. <nav>
- C. <section>
- D. <header>

Snippet 4: XML Syntax

- A. <note><to>User</to><from>Admin</from>
- B. <note><to>User</to><from>Admin</from></note>
- C. <note><to>User</from></note>
- D. <note to="User"></note>

Snippet 5: Web Communication Protocols

- A. FTP
- B. HTTP
- C. SMTP
- D. SNMP

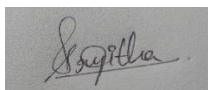
Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

Code of Conduct:

I s.srijitha(192471024)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

1. Select the correct HTML tag for video embedding and justify your answer.

Answer:

Correct Option: B

`<video src="video.mp4" controls></video>`

Explanation:

- The `<video>` tag is the standard HTML5 element used to embed video files on a webpage.
 - It supports video formats such as MP4, WebM, and Ogg.
 - The controls attribute displays built-in controls like Play, Pause, Volume, and Full Screen.
 - It improves user interaction by allowing users to control video playback.
 - The `<video>` element is supported by all modern web browsers.
 - Option A (`<media>`), Option C (`<movie>`), and Option D (`<vid>`) are not valid HTML5 elements.
 - Therefore, `<video>` is the correct and recommended tag for embedding videos in HTML5.
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2. Identify the appropriate HTTP method for form submission and explain why.

Answer:

Correct Option: B. POST

Explanation:

- The POST method is used to send data from the client to the server securely.
 - It sends the form data inside the HTTP request body instead of the URL.
 - Sensitive information such as usernames, passwords, and personal details is hidden from the browser's address bar.
 - POST is commonly used for registration forms, login forms, online payments, and file uploads.
 - Unlike GET, POST has no practical limit on the amount of data that can be sent.
 - GET is mainly used to retrieve information from the server.
 - PUT is used to update existing resources, while DELETE removes resources.
 - Hence, POST is the most suitable method for secure form submission.
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3. Choose the proper semantic element for navigation and justify its usage.

Answer:

Correct Option: B

`<nav>`

Explanation:

- The `<nav>` element is a semantic HTML5 tag that defines a navigation section of a webpage.
 - It usually contains important navigation links such as Home, About, Services, Contact, and Login.
 - Using `<nav>` improves the structure and readability of HTML documents.
 - Search engines can easily identify navigation links, which improves Search Engine Optimization (SEO).
 - Screen readers also recognize `<nav>`, making websites more accessible for visually impaired users.
 - The `<div>` tag is only a generic container without semantic meaning.
 - `<section>` is used to group related content, and `<header>` represents introductory content.
 - Therefore, `<nav>` is the correct semantic element for navigation menus.
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4. Identify the correct XML syntax and explain the error in other options.

Answer:

Correct Option: B

`<note>`

`<to>User</to>`

`<from>Admin</from>`

`</note>`

Explanation:

- XML requires every opening tag to have a matching closing tag.
 - Elements must be properly nested and enclosed within a single root element.
 - Option B follows all XML syntax rules correctly.
 - Option A is incorrect because the closing `</note>` tag is missing.
 - Option C contains mismatched closing tags (`</from>` is used instead of `</to>`).
 - Option D is invalid because the closing `>` is missing in the `</note>` tag.
 - XML is case-sensitive, and syntax errors prevent XML documents from being parsed correctly.
 - Therefore, Option B is the only valid XML document.
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5. Choose the correct protocol for web communication and justify your choice.

Answer:

Correct Option: B. HTTP

Explanation:

- HTTP (HyperText Transfer Protocol) is the standard communication protocol used between web browsers and web servers.
 - It allows browsers to request web pages, images, videos, and other resources from a server.
 - HTTP follows the request-response model, where the client sends a request and the server returns a response.
 - Most websites today use HTTPS, which is the secure version of HTTP with encryption.
 - FTP is used for transferring files between computers.
 - SMTP is used for sending emails.
 - SNMP is used for monitoring and managing network devices.
 - Therefore, HTTP is the correct protocol for web communication because it enables browsers and servers to exchange web content efficiently.
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